

Module - 2SemiconductorIntrinsic Semiconductors

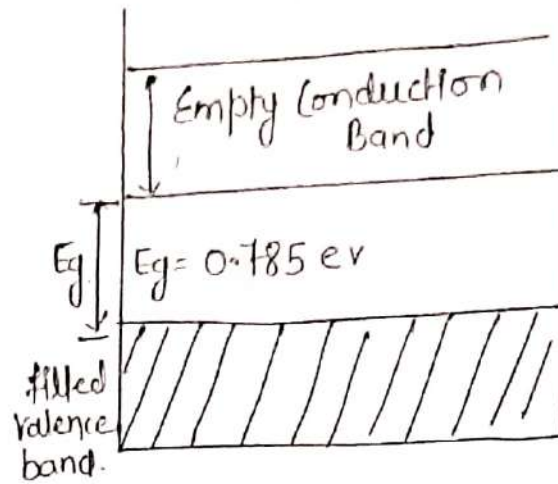
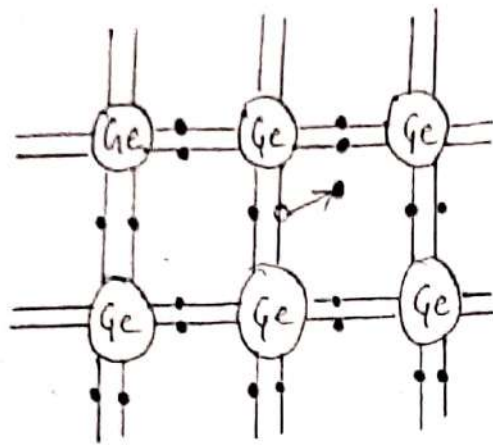
A pure Semiconductor which is free of every impurity is called Intrinsic Semiconductor. Electrical Conductivity of pure Semiconductor is totally due to number of electrons excited from valence band to Conduction Band.

Example :- Silicon & Germanium.

Silicon (atomic no. - 14)  $1s^2 2s^2 2p^6 3s^2 3p^2$

Germanium (atomic no. - 32)  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^2$

Both the atoms have four valence electrons. Crystal structure of germanium is shown below. The four valence electrons of germanium atom forms four covalent bond by sharing the electrons from neighbouring four germanium atom. Each covalent bond shares two electrons one from each atom. Hence outermost orbit of each Ge atom is complete with eight electrons.



- Even at room temp. thermal vibrations of the atom provide Energy to  $e^-$  so that these  $e^-$  jump from valence band to conduction band leaving hole in valence band
- In Intrinsic Semiconductor.

$$\boxed{n_e = n_h = n_i}$$

where  $n_e$  = density  $e^-$  in conduction band

$n_h$  = density of holes in valence band

$n_i$  = Intrinsic carrier density

- Here Concentration of holes &  $e^-$ 's are equal but  $\frac{2}{3}$  of current is due to  $e^-$  &  $\frac{1}{3}$  by holes because  $e^-$ 's are much more mobile than holes

- Resistivity decreases ( $\downarrow$ ) & conductivity ( $\uparrow$ ) increases with rise in temperature.



\* Conductivity of Semiconductor depends upon

- Concentration of Mobile charge carriers (holes &  $e^-$ )
- Mobility of charge carriers

When electric field is applied, Current density due to motion of  $e^-$  & holes are

$$J_n = en v_n \quad \text{A/m}^2$$

$$J_p = ep v_p \quad \text{A/m}^2$$

where  $n$  = density of free  $e^-$

$p$  = density of holes

$v_n$  = drift velocity of  $e^-$

$v_p$  = drift velocity of holes

→ Conductivity due to  $e^-$  is

$$\sigma_n = \frac{J_n}{E} = \frac{en v_n}{E} = en \mu_n$$

→ Conductivity due to holes

$$\sigma_p = \frac{J_p}{E} = \frac{ep v_p}{E} = ep \mu_p$$

where  $\mu_n$  &  $\mu_p$  are mobility of  $e^-$  & holes

→ Now, Total Conductivity  $\sigma = \sigma_n + \sigma_p$

$$\sigma = en \mu_n + ep \mu_p$$

$$\boxed{\sigma = e(n \mu_n + p \mu_p)}$$

In Case of Intrinsic Semiconductors

$$n = p = n_i$$

So, Conductivity of Intrinsic Semiconductor is

$$\sigma_i = e n_i (\mu_n + \mu_p)$$

## Extrinsic Semiconductor

At room temperature Intrinsic semi-conductor has little Current Conduction Capability.

In Order to use the Semiconductor in electronic devices, its Conduction properties should be increased by adding some impurities in the process of Crystallization. The added impurities are very small of the Order of one atom per million atom of pure Semiconductor.

Doping  $\Rightarrow$  The process by which controlled impurities are added to pure Semiconductors.

### Type of Extrinsic Semiconductor

N type Semiconductor

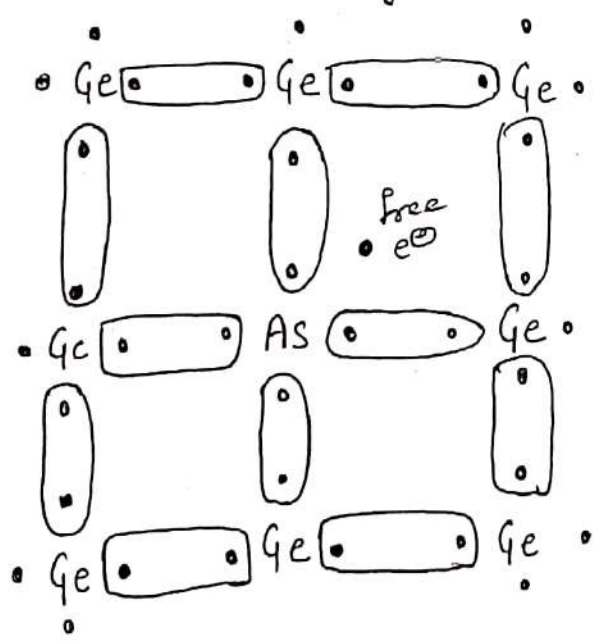
P type Semiconductor

## N type Semiconductor

→ when a small amount of Pentavalent impurities are added to pure Semiconductor.

Example:- Arsenic ( $Z=33$ )  
Antimony ( $Z=51$ )

→ act as donor impurities  
→ when Pentavalent arsenic is added to pure germanium its four valence  $e^-$  form covalent bond with four Germanium atom & fifth  $e^-$  remains free



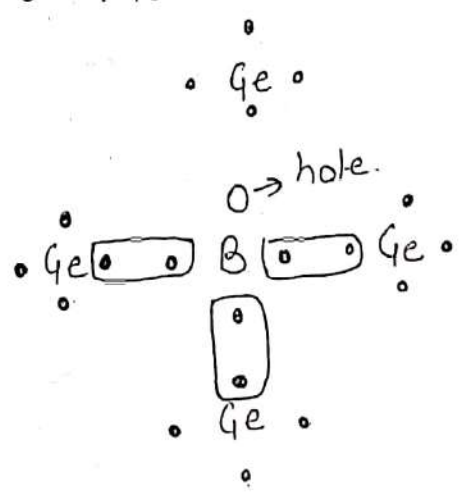
→ Majority Carriers are  $e^-$

## P type Semiconductor

when small amount of trivalent impurities are added to pure Semiconductor.

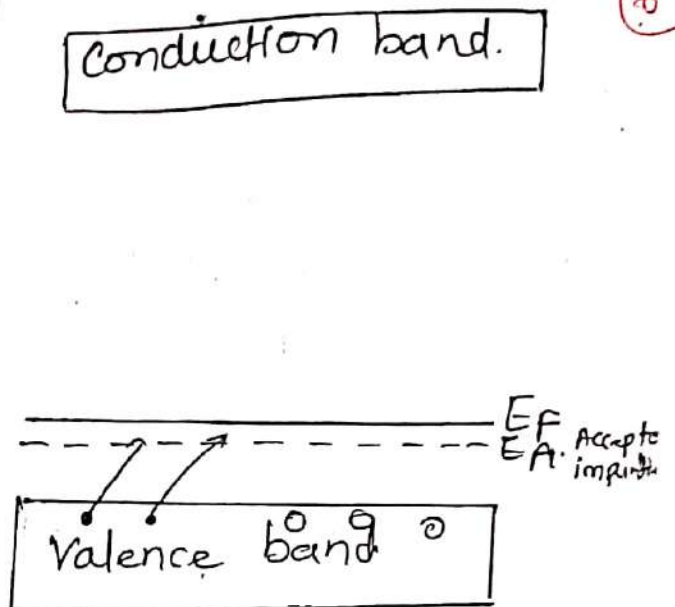
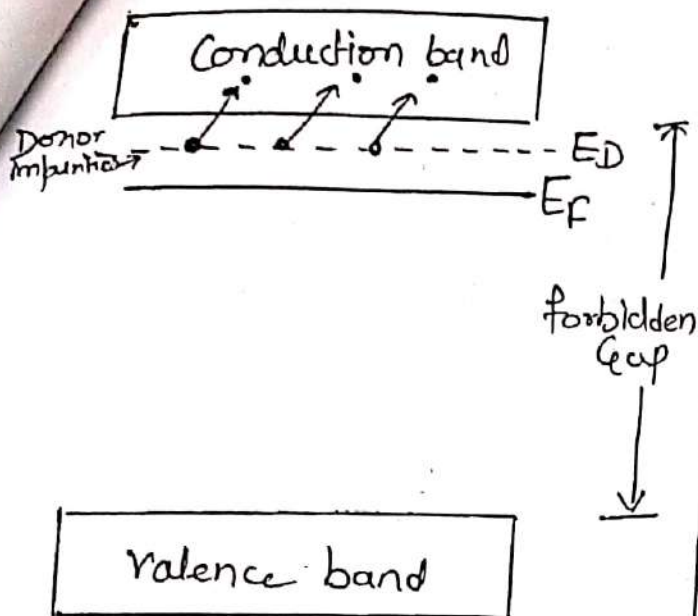
Example:- Boron, Gallium

→ act as acceptor impurities  
→ Three covalent bonds are there. whereas fourth bond is incomplete that is missing of  $e^-$  known as hole



→ Majority Carriers are holes





- Conductivity of N type is double that of P type as  $e^-$  are much more mobile than holes.
- N type & P type Semiconductors are electrically neutral.

# Number of Electrons in Conduction Band

Let us consider electrons in Conduction Band behaves as free particles with effective mass  $m_e^*$  & number of Conduction electrons ( $dn$ ) whose energies lies bt/w  $E$  &  $E+dE$  is

$$dn = N(E) f(E) dE \quad \text{--- (1)}$$

where  $N(E)$  is density of states of Conduction Band is given by

$$N(E) = \frac{4\pi}{h^3} (2m_e^*)^{3/2} (E - E_c)^{1/2} \quad \text{--- (2)}$$

where  $E > E_c$

$E_c$  = energy at the bottom of Conduction band

and  $f(E)$  = fermi Dirac distribution function

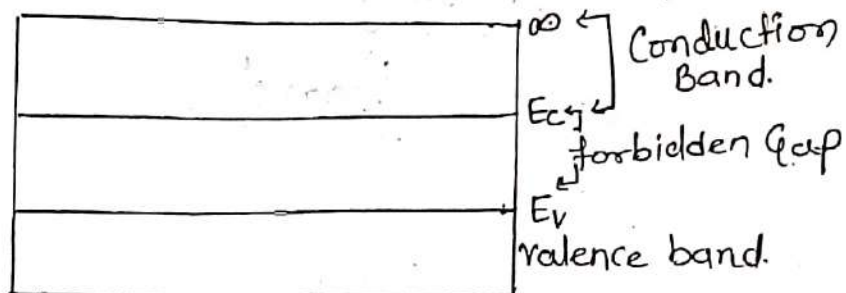
$$f(E) = \frac{1}{1 + e^{(E - E_f)/k_B T}} \quad \text{--- (3)}$$

where  $k_B$  = Boltzmann const  
 $T$  = Temp.

Now, electrons in Conduction band having energy bt/w  $E_c$  to  $\infty$ .

So, Concentration of electrons in Conduction band is given by.

$$n = \int_{E_c}^{\infty} N(E) f(E) dE \quad \text{--- (4)}$$



Now put the value of  $N(E)$  &  $f(E)$  using Eq (2) & (3) in (4)

$$n = \int_{E_c}^{\infty} \frac{4\pi}{h^3} (2m_e^*)^{3/2} \frac{(E-E_c)^{1/2}}{1 + e^{(E-E_F)/k_B T}} dE$$

as  $E-E_F \gg k_B T$ ; so neglected  $1$  in denominator

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} (E-E_c)^{1/2} e^{-(E-E_F)/k_B T} dE$$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} (E-E_c)^{1/2} e^{(E_F-E_c)/k_B T} dE$$

Now, dividing numerator & Denominator by  $(k_B T)^{1/2}$  & adding & subtracting  $E_c$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \underbrace{(k_B T)^{1/2}} \int_{E_c}^{\infty} \underbrace{\left( \frac{E-E_c}{k_B T} \right)^{1/2}} e^{(E_F - \underline{E_c} + \underline{E_c} - E)/k_B T} dE$$



$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} (k_B T)^{1/2} e^{(E_F - E_c)/k_B T} \int_{E_c}^{\infty} \left(\frac{E - E_c}{k_B T}\right)^{1/2} e^{-(E - E_c)/k_B T} dE \quad (5)$$

Let  $\frac{E - E_c}{k_B T} = x$  or  $\frac{dE}{k_B T} = dx$ .

So,  $dE = (k_B T) dx$

Now these values in above Eq<sup>n</sup> (5) we get

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} (k_B T)^{1/2} e^{(E_F - E_c)/k_B T} \int_0^{\infty} x^{1/2} e^{-x} (k_B T) dx$$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} (k_B T)^{3/2} e^{(E_F - E_c)/k_B T} \int_0^{\infty} x^{1/2} e^{-x} dx$$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} (k_B T)^{3/2} e^{(E_F - E_c)/k_B T} \left(\frac{\sqrt{\pi}}{2}\right)$$

(because  $\int_0^{\infty} x^{1/2} e^{-x} dx = \frac{\sqrt{\pi}}{2}$ )

$$n = 2 \left[ \frac{2\pi m_e^* k_B T}{h^2} \right]^{3/2} e^{(E_F - E_c)/k_B T} \Rightarrow \text{This is concentration (density) of electrons in conduction band in intrinsic semiconductor}$$

or  $n = N_c e^{-(E_c - E_F)/k_B T}$

where  $N_c = 2 \left[ \frac{2\pi m_e^* k_B T}{h^2} \right]^{3/2}$

# Number of Holes in Valence Band

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Hole means vacancy of electron

$f(E) \Rightarrow$  Means occupational probability for electrons

So, for Holes probability function becomes

$$1 - f(E) = 1 - \frac{1}{1 + e^{(E - E_F)/k_B T}}$$

$$= 1 - [1 + e^{(E - E_F)/k_B T}]^{-1}$$

$$= 1 - [1 - e^{(E - E_F)/k_B T}] \quad \left\{ \text{using Binomial expansion} \right\}$$

$$= e^{(E - E_F)/k_B T} \quad \text{--- (1)}$$

for top of valence band (maximum energy) density of states is

$$N(E) = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (E_v - E)^{1/2} \quad \text{--- (2)}$$

where  $m_p^* \Rightarrow$  is effective.

mass of hole near top of valence band

Now, number of Holes is

$$P = \int_{-\infty}^{E_v} N(E) [1 - f(E)] dE \quad \text{--- (3)}$$

|                  |           |
|------------------|-----------|
| Conduction Band. | $+\infty$ |
| forbidden Gap    | $E_c$     |
| Valence band.    | $E_v$     |
|                  | $-\infty$ |

Now, put the value of  $N(E)$  &  $[1 - f(E)]$  from Eq<sup>n</sup> (1) & (2) in Eq<sup>n</sup> (3)

$$P = \int_{-\infty}^{E_v} \frac{4\pi}{h^3} (2m_p^*)^{3/2} (E_v - E)^{1/2} e^{(E - E_F)/k_B T} dE$$

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} \int_{-\infty}^{E_v} (E_v - E)^{1/2} e^{(E - E_F)/k_B T} dE$$

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} \int_{-\infty}^{E_v} \left( \frac{E_v - E}{k_B T} \right)^{1/2} e^{(E - E_F + E_v - E_v)/k_B T} dE$$

( $\because$  dividing numerator & denominator by  $(k_B T)^{1/2}$  & add & subtracting  $E_v$ )

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_F)/k_B T} \int_{-\infty}^{E_v} \left( \frac{E_v - E}{k_B T} \right)^{1/2} e^{(E - E_v)/k_B T} dE \quad (4)$$

Let us put  $\frac{E_v - E}{k_B T} = x$ .

$$\frac{-dE}{k_B T} = dx$$

So,  $dE = -(k_B T) dx$ .



So Eq (4) become

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_f)/k_B T} \int_{-\infty}^0 x^{1/2} e^{-x} (-k_B T) dx.$$

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_f)/k_B T} \int_0^{\infty} x^{1/2} e^{-x} (k_B T) dx.$$

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_f)/k_B T} k_B T \int_0^{\infty} x^{1/2} e^{-x} dx.$$

$$\text{as } \int_0^{\infty} x^{1/2} e^{-x} dx = \frac{\sqrt{\pi}}{2}$$

So above Eq becomes

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_f)/k_B T} (k_B T) \frac{\sqrt{\pi}}{2}$$

$$P = 2 \left[ \frac{2\pi m_p^* k_B T}{h^2} \right]^{3/2} e^{(E_v - E_f)/k_B T}$$

⇒ This Eq is concentration of Holes in Valence band.

or  $P = N_v e^{(E_v - E_f)/k_B T}$

$$\text{where } N_v = 2 \left[ \frac{2\pi m_p^* k_B T}{h^2} \right]^{3/2}$$

# Fermi level in intrinsic Semiconductor

In intrinsic Semiconductor

No. of free electrons = No. of Holes

$$n = p$$

So 
$$N_c e^{-(E_c - E_f)/k_B T} = N_v e^{-(E_f - E_v)/k_B T}$$

Taking log both sides & rearranging terms

$$-\frac{(E_c - E_f)}{k_B T} = \log\left(\frac{N_v}{N_c}\right) - \frac{(E_f - E_v)}{k_B T}$$

Now, multiply both sides by  $k_B T$ .

$$-(E_c - E_f) = k_B T \log\left(\frac{N_v}{N_c}\right) - (E_f - E_v)$$

$$2E_f = k_B T \log\left(\frac{N_v}{N_c}\right) + (E_c + E_v)$$

$$E_f = \frac{E_c + E_v}{2} + \frac{1}{2} \log k_B T \log\left(\frac{N_v}{N_c}\right) \quad \text{--- (1)}$$

as 
$$N_v = 2 \left[ \frac{2 \pi m_p^* k_B T}{h^2} \right]^{3/2}$$

$$\& N_c = 2 \left[ \frac{2 \pi m_e^* k_B T}{h^2} \right]^{3/2}$$

Now, put the value of  $N_v$  &  $N_c$  in Eq<sup>n</sup> (1) we get

$$E_f = \left( \frac{E_c + E_v}{2} \right) + \frac{3}{2 \times 2} k_B T \log \left( \frac{m_p^*}{m_e^*} \right)$$

as intrinsic Semiconductors

$$m_p^* = m_e^*$$

So above Eq<sup>n</sup> becomes

$$E_f = \frac{E_c + E_v}{2}$$

Thus, fermi level in intrinsic Semiconductors lies exactly half way between top of Valence band & bottom of Conduction band. i.e. Centre of forbidden energy gap.

## Fermi level in Extrinsic Semiconductor

In Extrinsic Semiconductors, the number of  $e^-$  in Conduction band & number of Holes in Valence band are not equal. Hence, the probability of Occupation of energy level in Conduction band & Valence band are not equal. Therefore, the fermi level for



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extrinsic Semiconductor lies close to Conduction or Valence band

## Fermi level in N type Semiconductor

In N type Semiconductor pentavalent impurities are added. Each pentavalent impurity donates free  $e^-$ . The addition of pentavalent impurity creates large number of free  $e^-$  in Conduction band.

At room temp., the number of  $e^-$  in Conduction band is greater than number of Holes in Valence band. Hence, the probability of Occupation of energy levels by electrons in Conduction band is greater than the probability of Occupation of energy levels by Holes in Valence band. This probability of Occupation of energy levels is represented in terms of Fermi level. Therefore, the Fermi level in n type Semiconductor lies close to the Conduction band.

Fermi level for N type Semiconductor is

$$E_F = E_c - k_B T \log \frac{N_c}{N_D}$$

Where  $E_f$  = Fermi level

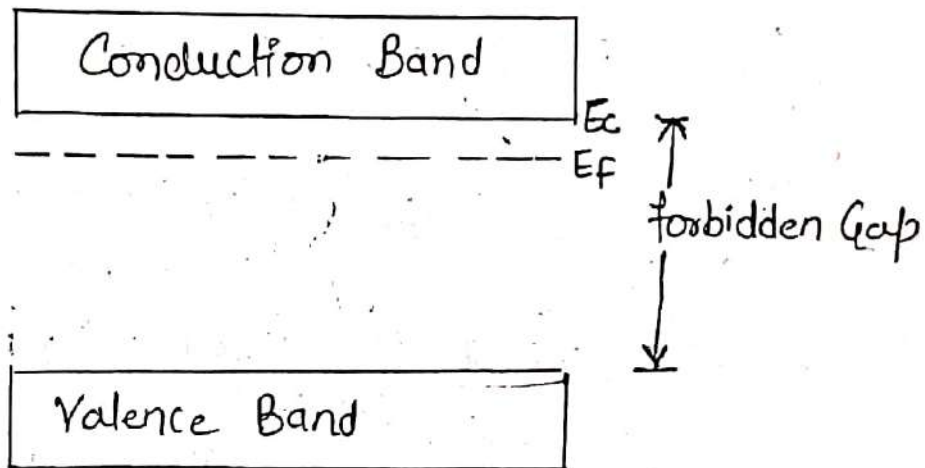
$E_c$  = Conduction Band

$k_B$  = Boltzmann Constant

$T$  = absolute Temp.

$N_c$  = density of states in Conduction band

$N_D$  = Concentration of donor atoms.



## Fermi level in p-type Semiconductor

In p type Semiconductor trivalent impurity is added. Each trivalent impurity creates a hole in valence band & ready to accept an electron. The addition of trivalent impurity creates large number of holes in valence band.

At room temp. number of holes in

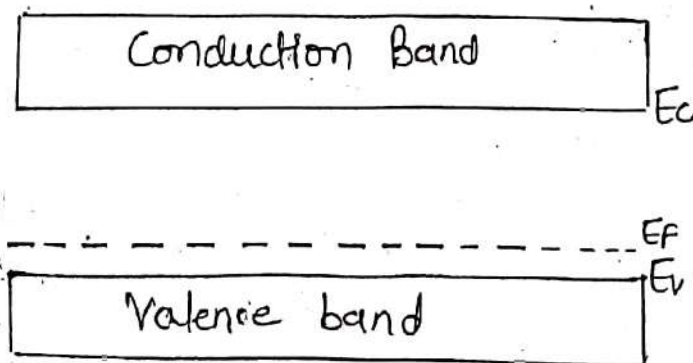
Valence band is greater than the number of electrons in conduction band. Hence, the probability of occupation of energy levels by holes in valence band is greater than probability of occupation of energy levels by electrons in conduction band.

This probability of occupation of energy level is represented by Fermi level. Therefore, the Fermi level in p type semiconductor lies close to valence band & is given as

$$E_F = E_v + k_B T \log \frac{N_v}{N_A}$$

where  $N_v$  = density of states in valence band

$N_A$  = concentration of acceptor atoms.





# Effect of Temperature on fermi level

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## (1) N Type Semiconductors

For n type semiconductor number of electrons is more than number of holes. That is mostly because there are additional electron being supplied by donor atoms (pentavalent atom). So let us consider

$$n \approx N_D \quad \text{where } N_D = \text{Concentration of Donor atoms}$$

as we know Carrier Concentration in N type semiconductor is

$$n = N_C e^{-(E_C - E_F)/k_B T}$$

$$\text{So } N_D = N_C e^{-(E_C - E_F)/k_B T} \quad \text{where } N_C = 2 \left[ \frac{2 \pi m_e^* k_B T}{h^2} \right]^{3/2}$$

$$\frac{N_C}{N_D} = e^{(E_C - E_F)/k_B T}$$

$$\log \left( \frac{N_C}{N_D} \right) = \frac{E_C - E_F}{k_B T} \quad (\text{Taking log both sides})$$

$$k_B T \log \left( \frac{N_C}{N_D} \right) = E_C - E_F$$

$E_F = E_C - k_B T \log \left( \frac{N_C}{N_D} \right)$

 $\Rightarrow$  This Eq<sup>n</sup> shows that with rise in temp, fermi level drops

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(19)

This Eq<sup>n</sup> shows that rise in temperature, Fermi level drops because at higher temperature. Semiconductor will produce more electron hole pair. Due to this production of extra  $e^-$  hole pair Fermi level moves down toward the valence band.

## (2) P type Semiconductor

In p type Semiconductor Concentration of holes can be approximately equal to

$$p \approx N_A \quad \text{where } N_A = \text{Concentration of acceptor atom (trivalent atom)}$$

We know, for carrier concentration in p type Semiconductor.

$$p = N_V e^{-(E_F - E_V)/k_B T} \quad \text{--- (1)}$$

where  $N_V = 2 \left[ \frac{2\pi m_p^* k_B T}{h^2} \right]^{3/2}$  called density of states in valence band.

So Eq<sup>n</sup> (1) becomes

$$N_A = N_V e^{-(E_F - E_V)/k_B T}$$

$$\frac{N_V}{N_A} = e^{(E_F - E_V)/k_B T}$$

$$\log \left( \frac{N_V}{N_A} \right) = \frac{(E_F - E_V)}{k_B T}$$

$$k_B T \log \left( \frac{N_V}{N_A} \right) = E_F - E_V$$

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$$\boxed{E_F = E_V + k_B T \log \left( \frac{N_V}{N_A} \right)}$$

⇒ This Eq<sup>n</sup> shows that with rise in temp Fermi level rise toward Conduction band.

Because more temperature means more Energy of electron tends to move towards the Conduction band, which push the Fermi level up.



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So, product of product of electron & hole densities is constant i.e.

$$\boxed{n p = n_i^2} \text{ at Equilibrium}$$

Carrier Generation & recombination process is fundamental to operation of many optoelectronic semiconductor devices such as photodiodes, LED & laser diodes.

### Carrier Transport:- Drift & Diffusion

There are two basic transport mechanisms

drift:- Movement of charge carriers due to electric field

diffusion:- flow of charges due to density gradient

Drift Current:- When electric field is applied to the semiconductor, it produces a force on electrons & holes so that they experience a net acceleration & net movement. This net movement of charge carriers due to electric field is called drift. The net drift of charge gives rise to drift current.

So, drift current density due to motion of electrons in conduction band is

$$J_e = n e \mu_e E$$

where  $n$  = no. of free  $e^-$   
 $\mu_e$  = mobility of  $e^-$   
 $E$  = Electric field applied

Drift current due to motion of holes in Valence band

$$J_h = p e \mu_h E$$

So, total current density.  $J = J_e + J_h$

$$J = n e \mu_e E + p e \mu_h E$$

$$J = e E (n \mu_e + p \mu_h)$$

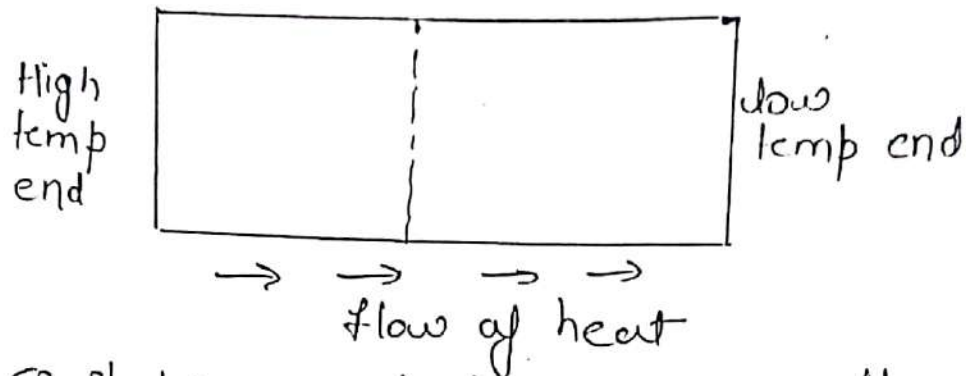
Note:- Drift current flows only when external electric field is applied across the semiconductor

## Diffusion Current

In case of semiconductors, current can also be flow without the application of external electric field.

Let us consider the material having two different temperatures across

its two ends. Due to temperature gradient heat flows from higher temperature region to lower temperature region until the temperature is uniform throughout the material.



Similarly, an electric current flow in Semiconductor even in the absence of applied Electric field provided a concentration gradient that exist in the material.

A concentration gradient exist if the number of electrons or holes is greater in one region of Semiconductor as compare to rest of region

In Semiconductors charge carriers have tendency to move from the region of higher concentration to lower concentration. Thus the movement of charge carriers takes place resulting in current called diffusion current.



Diffusion Current density due to flow of electrons

$$J_e (\text{diff}) = e n_e \frac{dn}{dx}$$

Diffusion Current density due to flow of holes

$$J_h (\text{diff}) = e n_h \frac{dp}{dx}$$

## Difference between

### Diffusion Current

→ Diffusion Current may occur even electric field not applied to Semiconductor.

→ Magnitude of diffusion current depends of the slope of carrier concentration

→ Diffusion Current flow in the direction of slope of carrier concentration & does not obey ohm's law

### Drift Current

Drift Current required the presence of electric field.

Drift Current depends on the concentration of Semiconductor material

Drift Current flows in the direction of electric field & depends on Ohm's law

# P-N Junction

P-N Junction is formed by diffusing P type material to one half side & N type material to other half side. The plane dividing the two zones is known as junction.

P-type & N-type semiconductor pieces before joining is shown in fig.

In figure, P-type semiconductor has negative acceptor ion (shown by encircled minus sign) & positive charge holes. Similarly, in N-type semiconductor has positive donor ions (shown by positive sign) & negative charged free electrons.

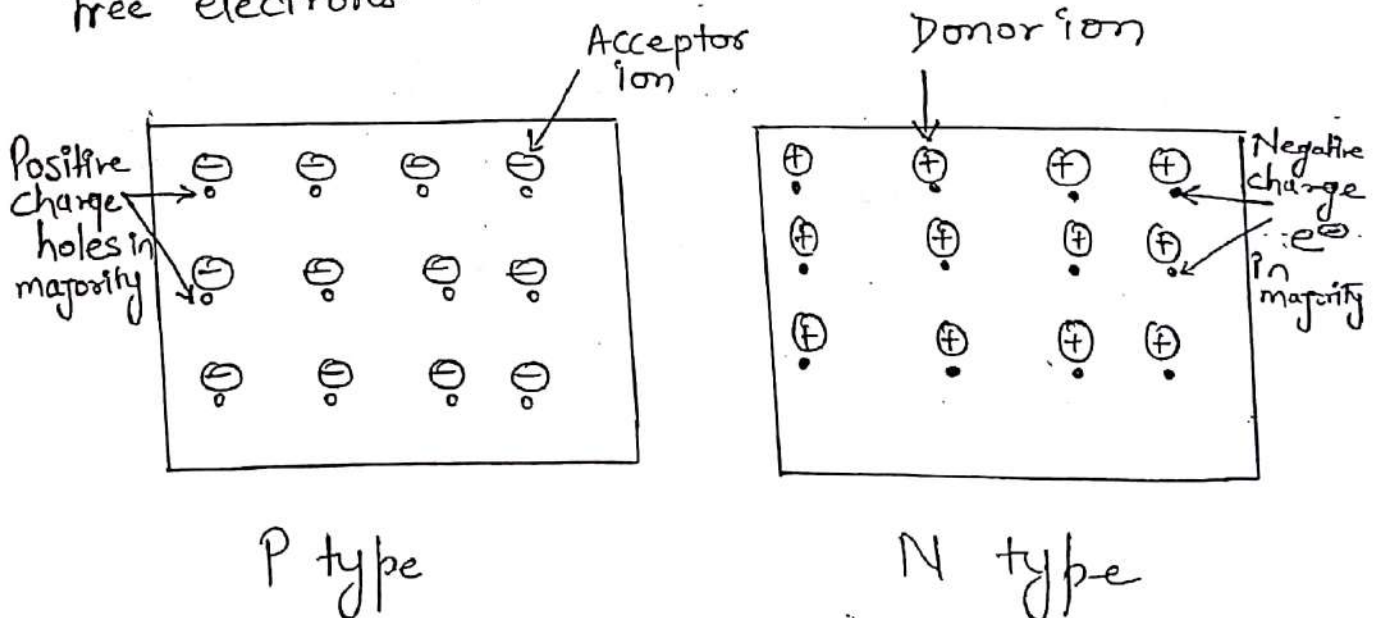


fig. P-type & N-type Semiconductor before joining.

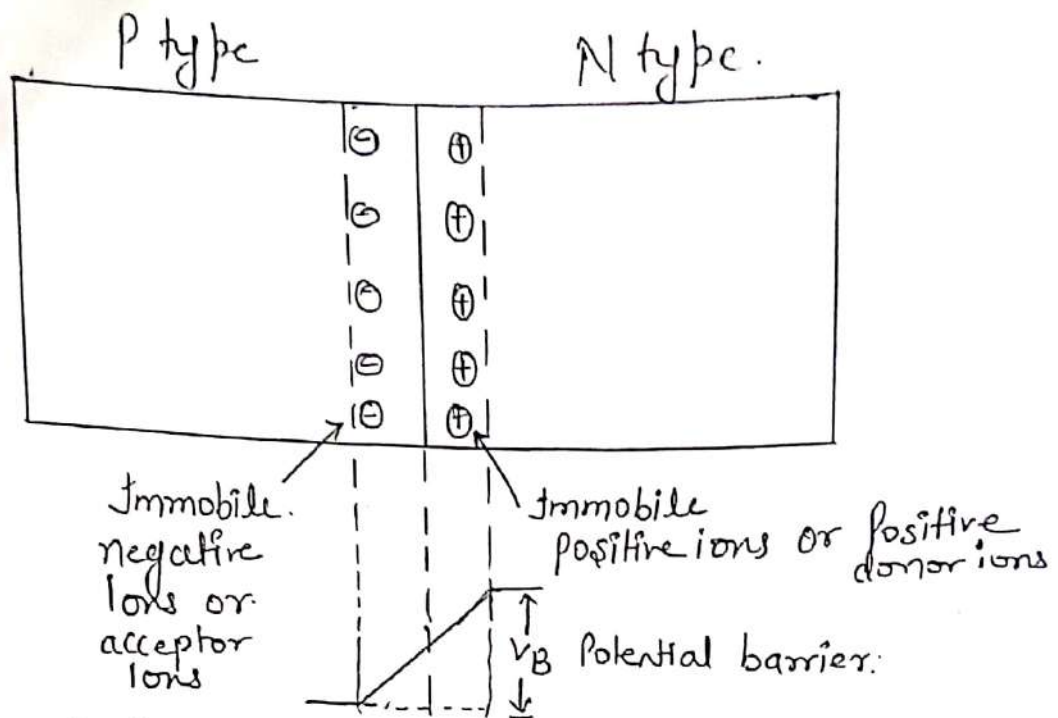
## Formation of depletion Layer

Now, Consider that two pieces are joined together as shown in fig. below. As p type material has high concentration of holes & n type material has high concentration of electrons hence there is tendency of holes to diffuse over n-side & electrons to p-side. This process is called diffusion. So due to diffusion some of holes from p-side cross over to n-side & become neutral. Similarly, some of electrons from n-side cross over to p-side where they combine with holes & become neutral.

Thus, a region is formed known as "depletion region" or space charge region.

The diffusion of holes & electrons continues till potential barrier in space charge region which prevent further diffusion or neutralization.





### Some important points

⇒ When P-N junction is formed, the holes from P region diffuse to N-region where they disappear due to recombination with electrons

⇒ The electrons from N region diffuse to P region where they disappear due to recombination with holes

⇒ In this procedure, negative immobile acceptor ions in P-region & positive immobile donor ions in N-region are left near junction

⇒ Now, no further diffusion of holes & electrons takes place across the junction

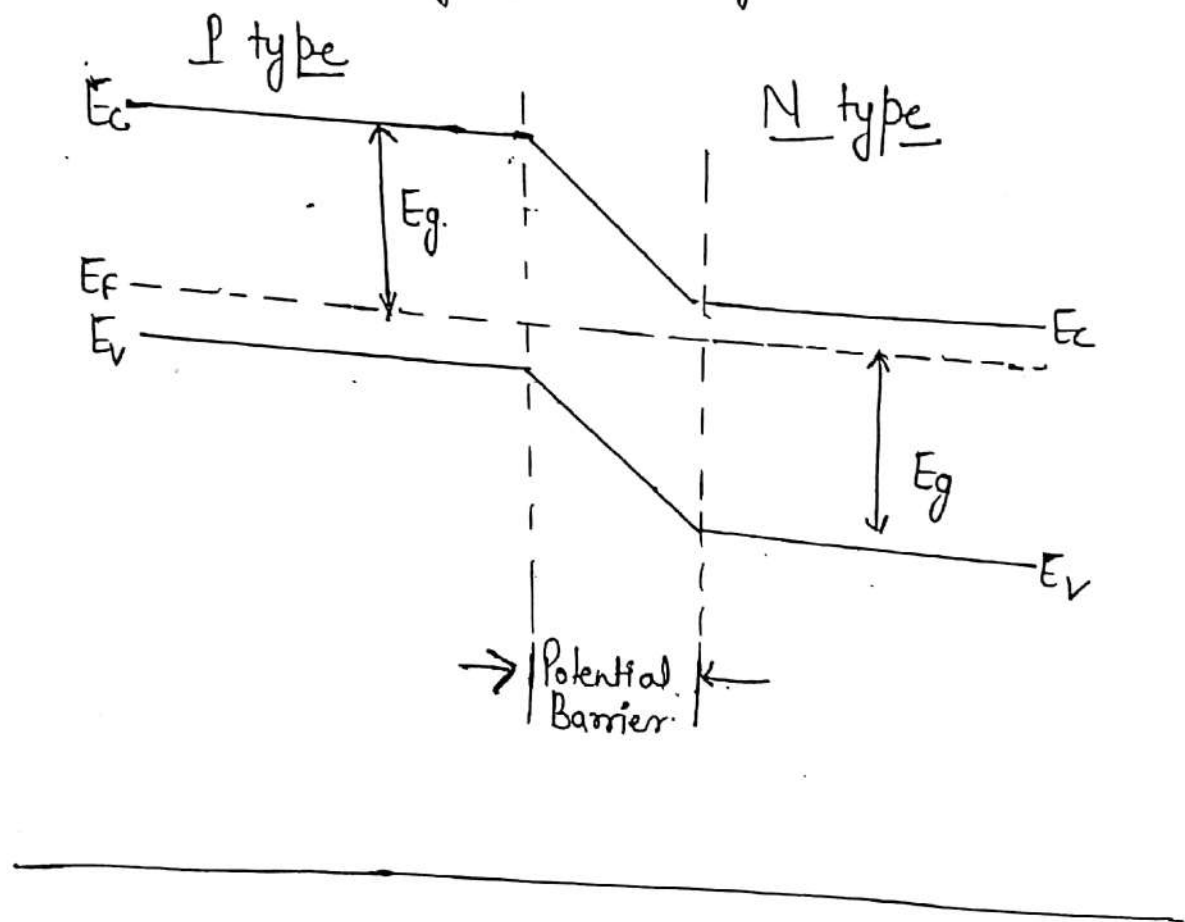
because holes trying to diffuse in N region are repelled by immobile positive ions & the electrons trying to diffuse in P-region are repelled by immobile negative ions

⇒ The region containing immobile negative & immobile positive ions is called depletion region.

⇒ As negative ions (immobile) are created on P-side of junction so P-type acquire negative potential. Similarly, positive ions are created on N-side so N side acquire positive potential. Hence the voltage " $V_B$ " is developed across the junction under equilibrium condition. This voltage is known as Junction Voltage or Internal voltage or potential barrier voltage " $V_B$ " which is about 0.3 V for Ge & 0.7 V for Si at room temp. 300 K

⇒ we know Fermi level lies near the valence band in P type & near the conduction band in N type Semiconductors under thermal equilibrium, the

Average energy of p & n regions are become equal so fermi level ( $E_F$ ) of p & n regions will be at same level as shown in fig. below. The equilization of fermi level in p & n type is analogous to equilization of level of water in two containers on being joined together.



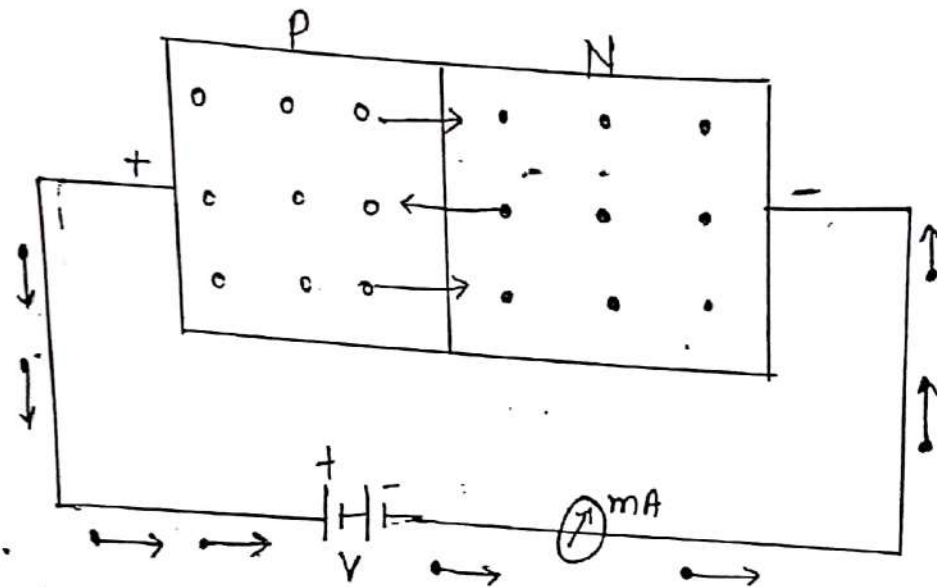


# P-N Junction Diode

1.

## Forward Bias

When an external voltage is applied to P-N Junction in such a direction that it cancel the potential barrier & permits the current flow is called forward bias.



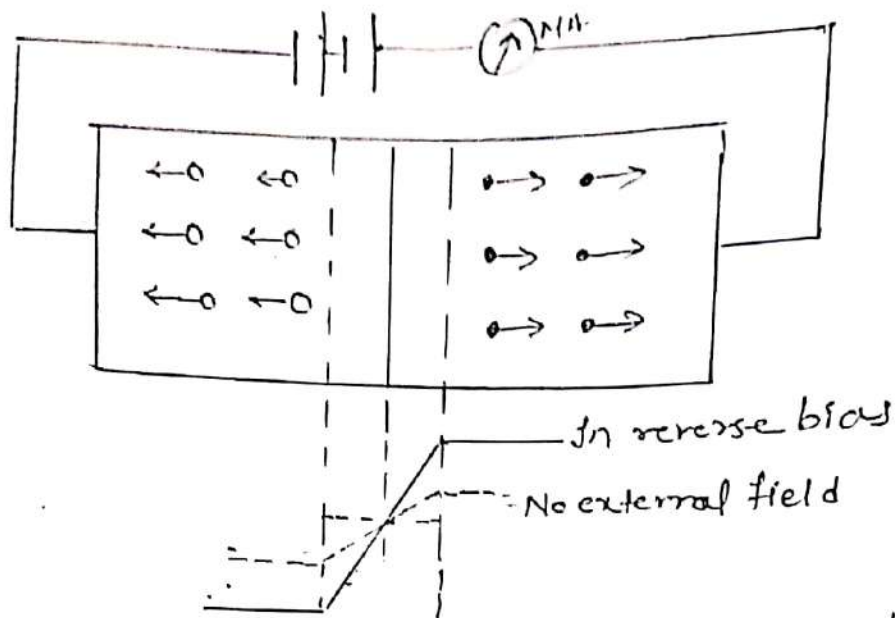
In forward bias

- (1) holes from P type Semiconductor is repelled by positive terminal of battery towards the junction. & simultaneously, the electron of N type Semiconductor are repelled by negative terminal toward junction.

2. Here battery voltage should be high to impart sufficient energy to these carriers to overcome the potential barrier & enable to cross through it.
3. When an electron hole recombine takes place near the junction, a covalent bond near positive terminal of battery breaks down. This causes liberation of electron which enters positive terminal of battery.
4. This action creates a new hole which moves towards the junction.
5. For each electron in N-region that combines with a hole from P region, an electron enters the crystal from negative terminal of the battery. The constant movement of electrons toward positive terminal of the holes towards the negative terminal produces a high forward current.

## 2. Reverse Bias

When external voltage is applied to PN Junction in such a direction that it increases the potential barrier then it is called reverse bias.



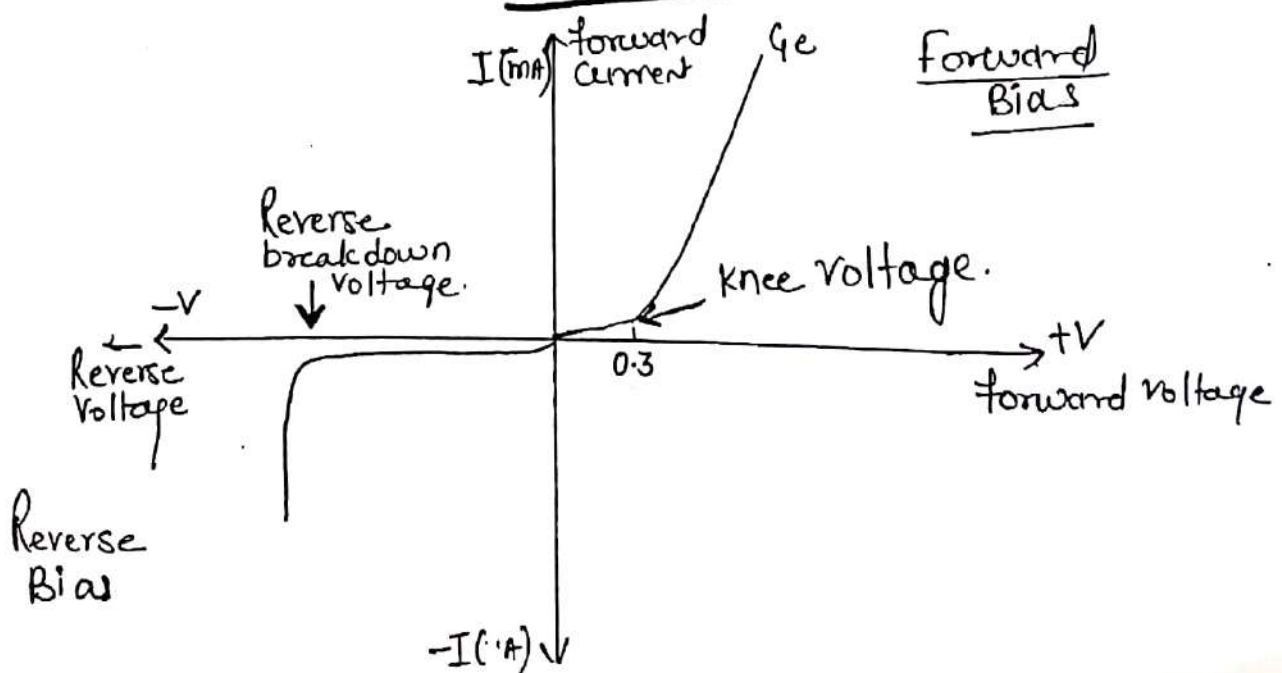
1. In reverse bias, reverse voltage establishes an electric field in the same direction of potential barrier. Therefore, the resultant field at junction is strengthened.  $\therefore$  barrier height is increased.
2. This increased potential barrier prevents the flow of charge carriers across the junction. Thus high resistance established.
3. In reverse bias electrons in N type & holes in p type semiconductor are attracted away from junction. Since there is no recombination of electron hole pair, so no current flow in circuit.



- following points are important
- (1) Junction potential barrier is strengthened.
  - (2) It offers high resistance
  - (3) There will be little current (in  $\mu A$ ) flows through the junction due to minority carriers.

Conclusions:- PN Junction diode is one way device which offers low resistance. when forward bias it behaves like an insulator when reverse bias. Thus, it can be used as rectifier i.e. for converting a.c in d.c

## V-I Characteristics of PN Junction



(1) In forward bias at forward voltage (0.3V for Ge + 0.7V for Si) the potential barrier across p-n junction is eliminated + current starts flow. This voltage is called "threshold voltage" or "knee voltage"

→ (2) As applied voltage further increases, the forward current rises exponentially as shown in Graph

→ (3) In reverse bias, as the reverse voltage is increased from zero, the reverse current quickly rises to its maximum or saturation value. The slightly increase is due to impurities on the surface of semiconductor. So current produced is called surface leakage current

→ (4) If reverse voltage is further increases kinetic energy of minority carriers becomes so high that they knock out the electron from atom. At this stage "junction breakdown" occur + there is sudden rise of reverse current.

# Reverse Bias characteristics Zener and Avalanche Breakdown

## ⇒ Avalanche Breakdown

The charge carriers of P-N junction diode sometime absorb the heat from the environment at normal room temperature. When reverse bias applied across the junction, the kinetic energy of electrons increases & they start moving with high velocity. While moving they collide with other atoms & creates number of free electrons which causes reverse saturation current. Because of this saturation current, the avalanche breakdown occur in diode.

## ⇒ Zener Breakdown

Zener breakdown takes place in heavily doped diodes. When high electric field is applied across the diode, then electron start moving across the junction. This develop small reverse bias current. but when movement of electron is increases



beyond the rated capacity of diode, then current rapidly increases. This is known as Zener breakdown

## Difference between

### Avalanche Breakdown

- Avalanche breakdown occurs because of collision of electrons
- It will destroy the junction
- Pair of electrons & holes are produced
- It occurs with low doping
- Breakdown voltage is directly proportional to temperature
- Temperature coefficient of avalanche breakdown is positive
- \* positive temp coefficient means temperature of material increases with reverse voltage

### Zener Breakdown

- Zener breakdown occurs because of high electric field
- It will not destroy the junction
- Electrons are produced
- It occurs with high doping
- Breakdown voltage is inversely proportional to temperature
- Temperature coefficient of Zener breakdown is negative
- \* negative temp coefficient means temperature decreases with reverse voltage.

# Diode Equation

In PN Junction forward bias, beyond knee voltage, forward current rises exponentially according to following equation

$$I = I_0 \left[ \exp\left(\frac{V}{\eta V_T}\right) - 1 \right] \quad \text{--- (1)}$$

Where  $I$  = diode current

$I_0$  = diode reverse saturation current at room temp

$V$  = applied forward voltage.

$\eta$  = constant [1 for Ge, 2 for Si]

$V_T = \frac{kT}{q} = \frac{T}{11,600}$  volt Equivalent to temp

$k$  = Boltzmann Const =  $1.3806 \times 10^{-23} \text{ J/K}$

$q$  = electronic charge =  $1.6 \times 10^{-19} \text{ C}$

$T$  = Temp (K)

In forward bias

$$\exp\left[\frac{V}{\eta V_T}\right] \gg 1$$

So Eq (1) becomes

$$I = I_0 \exp\left[\frac{V}{\eta V_T}\right]$$

Reverse Bias:- In Reverse bias, a constant reverse current saturation current  $I_0$  flow due to minority carriers.

Here  $V$  is negative so Eq<sup>n</sup> (1) can be modified as

$$I = I_0 \left[ \exp\left(-\frac{V}{\eta V_T}\right) - 1 \right]$$

In reverse region

$$\exp\left[-\frac{V}{\eta V_T}\right] \ll 1$$

so

$$I = -I_0$$

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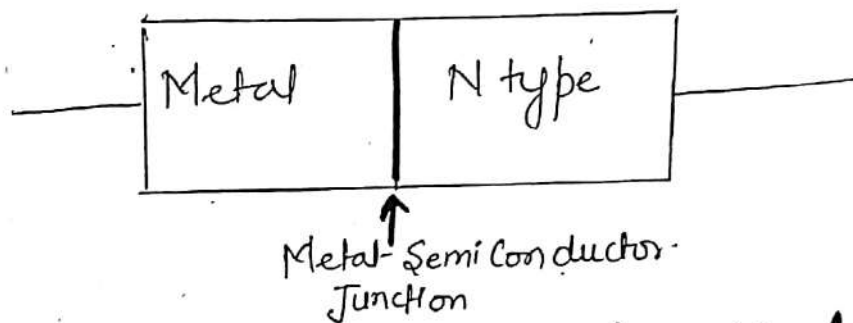
## Metal Semiconductor Junction or. Schottky Diodes

Schottky diode is a metal semiconductor junction diode that has less forward voltage drop than PN Junction & can be used in high speed switching applications.

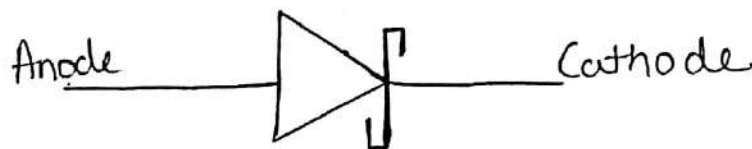


When p type & N type Semiconductors are joined then p-n junction is formed while in Schottky diode metals such as aluminium or platinum replace the p type semiconductor.

When aluminium or platinum metal is joined with N type semiconductor, a junction is formed between metal & N type semiconductor. This junction is known as metal semiconductor junction & depletion layer known as Schottky barrier.

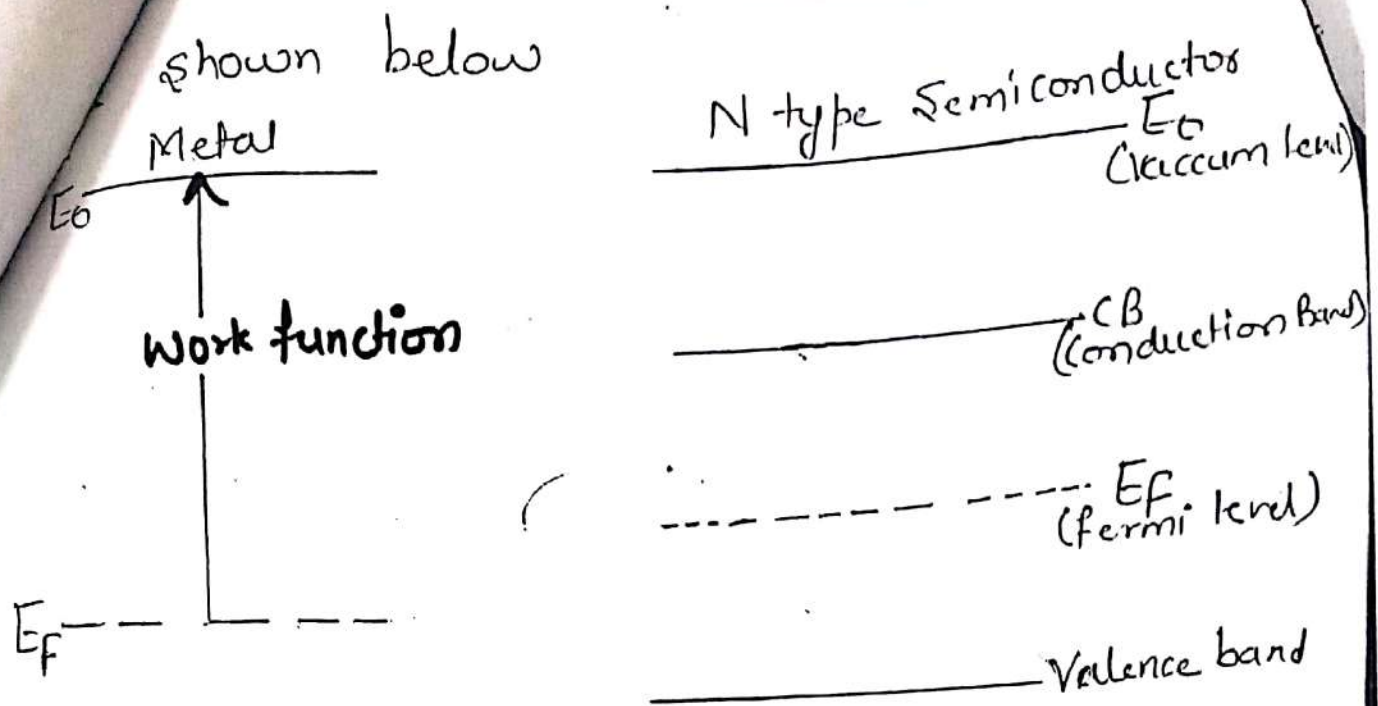


## Symbol of Schottky diode



## Energy Band diagram

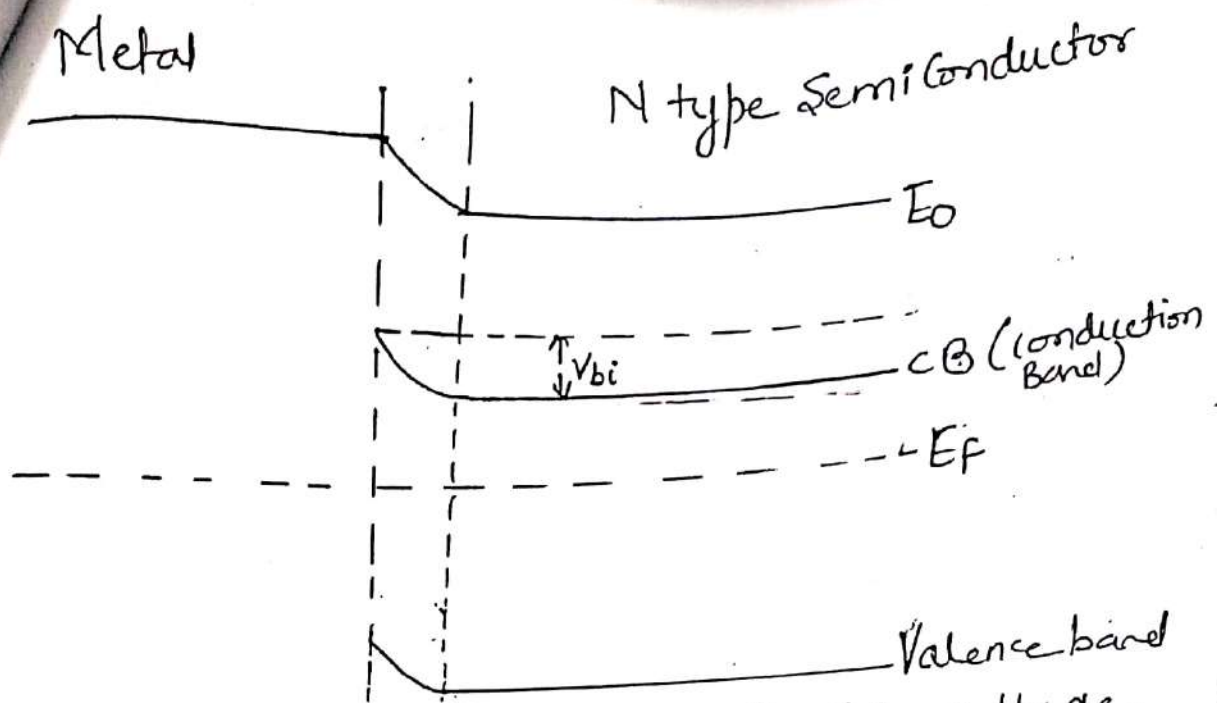
Energy band diagram of Schottky diode.



Vacuum level is the energy level of electrons that are outside the material. Work function is defined as energy required to move electron from Fermi level to vacuum level ( $E_0$ ).

The work function is different for metal & Semiconductor. Work function of a metal is greater than work function of Semiconductor. In Energy level diagram Fermi level at N-type Semiconductor lies above the metal side. We know electrons in higher energy level have more potential energy than electrons in lower energy level.

So the electrons in N-type Semiconductor have more potential energy than electrons in metals.



where  $V_{bi}$  = Built in voltage barrier

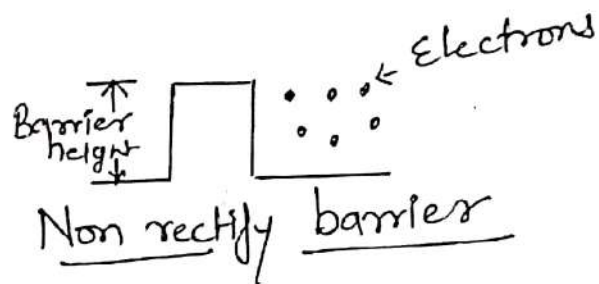
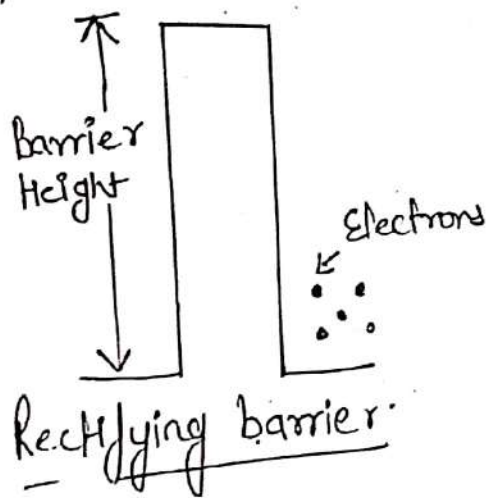
When metal is joined with N type Semiconductor then Schottky diode formed. The built in voltage ( $V_{bi}$ ) for Schottky diode is given by the difference b/w workfunction of metal & N type Semiconductor

### Some Important Concepts

- When Metal is in contact with lightly doped Semiconductor then rectifying Schottky diode (Non Ohmic Contact) formed while if metal is in contact with heavily doped Semiconductor then Non rectifying Schottky diode formed. (Ohmic Contact)
- Non rectifying metal Semiconductor (ohmic Contact) offer very low resistance than Rectifying metal Semiconductor junction (Non-Ohmic Contact)



→ In Non rectifying Schottky barrier, the barrier height is not enough to form depletion region. So depletion region is negligible or absent in ohmic contact diode while, in rectifying Schottky barrier, the barrier height is high enough to form depletion region. So depletion region is present in non ohmic junction diode



### Advantages:-

- Schottky diode can switch on & off much faster than PN junction diode
  - Schottky diode produce less unwanted noise than PN junction diode
  - Schottky diode has a less voltage drop (approx 0.2 to 0.3 volts) than PN junction (0.6 to 0.7 volts)
- Because of these advantages it is (Schottky diodes) used for high speed switching power circuits

- High Efficiency
- Schottky diodes operates at high frequencies

## Disadvantages

- Schottky diode produce large reverse saturation current than p-n junction diode.

## Semiconductor materials for optoelectronic devices

Si & Ge are the commonly used semiconductors. They belong to group IV of periodic table & used either in pure (intrinsic) or doped (extrinsic) form. The intrinsic & extrinsic semiconductors are known as "elemental semiconductors".

A compound semiconductor is a semiconductor composed of elements from two or more different groups of periodic table for eg:- one element from column III & one from column V, so called compound III-V semiconductor such as GaAs, InP & GaN.

The elemental semiconductors Si and Ge are indirect band gap semiconductors & therefore not suitable for producing light.

So they are basically used for making rectifying diodes while Compound Semiconductors are direct band gap Semiconductors & have high optoelectronic conversion capacity. due to which they are widely used in fabrication of optoelectronic devices such as semiconductor lasers, LED, photodiodes, solar cells etc.



## 2.11 Metal-semiconductor junctions

Development of electronic devices necessitates putting together two or more unlike materials (semiconductors, metals, insulators). The interface between these materials is known as the junction. So, three types of junctions can be formed:

- Metal-metal junction (used to make thermocouples)
- Metal-semiconductor junction (Schottky barrier or as an Ohmic contact)
- Semiconductor-semiconductor junction (p-n junction)

Metal-to-semiconductor junctions are of great importance as it affects the electrical properties (transport) of the devices. They can behave either as a Schottky barrier or as an Ohmic contact depending on the characteristics of the interface.

### 2.11.1 Schottky junctions

When a metal and semiconductor are brought into contact, there are two types of junctions formed depending on the work function of the semiconductor and its relation with the metal

1. Schottky junction -  $\phi_m > \phi_{semi}$
2. Ohmic junction -  $\phi_m < \phi_{semi}$

$\phi_{semi} < \phi_m$  fermi metal  
fermi semi 7

Let us consider a junction is formed involving a metal and n-type semiconductor, as shown in figure 6. Since the work function of the semiconductor is lower than that of metals so its Fermi level will be higher than that of metal. The Fermi levels must line up at equilibrium whenever a metal-semiconductor junction is formed. We know that there are electrons in the conduction level of the semiconductor which can move to the empty energy states above the Fermi level of the metal. So, there is a net positive charge on the semiconductor side and due to the excess electrons, a net negative charge is also there on the metal side which means there is an electric field which goes from the semiconductor to the metal leading to a contact potential as shown in figure 7. Now, in the case of a metal, the electron density is high i.e. of the order of  $10^{22} \text{ cm}^{-3}$  and is usually lower (typically  $10^{17} \text{ cm}^{-3}$ ) in the case of a semiconductor. So, when a contact is formed between two metals, the charges reside on the surface. On the other hand, when a contact is formed between a metal and semiconductor,

due to the low charge density on the semiconductor side, the electrons are removed not only from the surface.

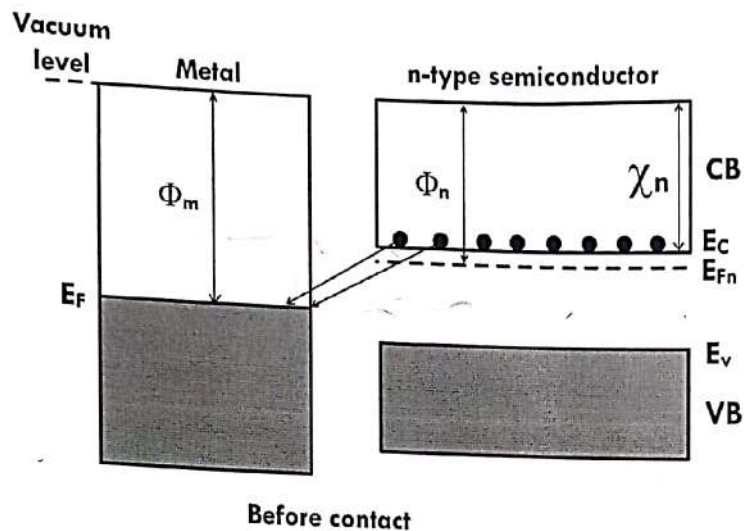


Figure 6: Schottky junction between metal and n-type semiconductor before contact. The work function of the semiconductor is smaller than the metal so that electrons can move from semiconductor to metal, forming a contact potential. Adapted from Principles of Electronic Materials - S.O. Kasap.

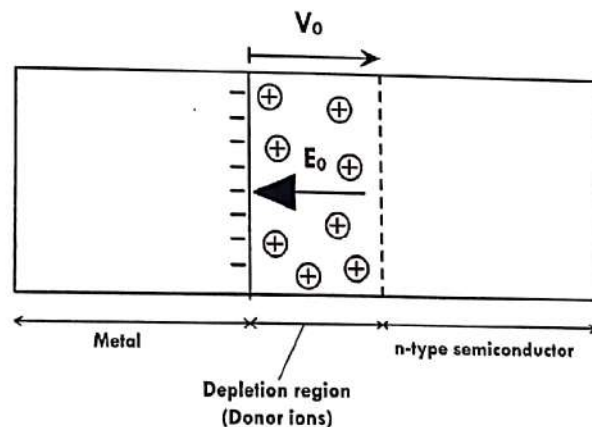


Figure 7: Schematic showing the metal, n-type semiconductor, and the Schottky junction between them. There is a depletion layer in the n-type semiconductor due to transfer of electrons to the metal. This leads to the formation of a contact potential. Adapted from Principles of Electronic Materials - S.O. Kasap.

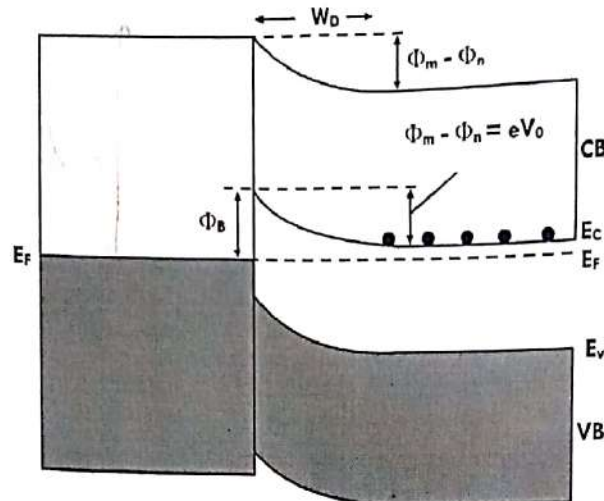


Figure 8: Schottky junction showing the band bending on the semiconductor side. Semiconductor bands bend up going from the semiconductor (positive) to metal (negative) since this is the same direction as the electric field. Adapted from Principles of Electronic Materials - S.O. Kasap.

but also penetrate at distance within the bulk of the semiconductor. So, there is a region within the semiconductor where electrons are lost as they move into the metal and this region is called the depletion region (also called space charge layer), shown in figure 7. Thus, when a Schottky junction is made between the metal and semiconductor, the Fermi level lines up and a positive potential is formed on the semiconductor side. Also, there is bending of the energy bands on the semiconductor side because of the depletion region which extends within a certain depth in the semiconductor. Bands bend up in the direction of the electric field (field goes from positive charge to negative charge, opposite of the potential direction). This means the energy bands bend up going from n-type semiconductor to metal, shown in figure 8. The Fermi levels line up and there is a certain region in the semiconductor (denoted by  $W$ ) where the bands bend (this is the depletion region). So, we now have an electric field and the contact potential ( $eV_0$ ) between the metal and the semiconductor and this contact potential depends upon the difference between the work functions

$$eV_0 = \phi_m - \phi_{semi} \quad (1)$$



Now, we know that the work function for the metal is a constant while in case of semiconductor, it depends on the dopant concentration (since this affects the Fermi level position). The contact potential then represents the barrier for the electrons to move from the n-type semiconductor to the metal. Initially, when the junction is formed, electrons move to the metal to create the depletion region. The contact potential thus formed prevents further motion of the electrons to the metal. There is also a barrier for electrons to move from metal to semiconductor. This is called the Schottky barrier and denoted by  $\phi_B$  in figure 8. This is given by

$$\phi_B = (\phi_m - \phi_n) + (E_c - E_{Fn}) = \phi_m - \chi_n \quad (2)$$

where  $\chi_n$  is the electron affinity of the n-type semi-conductor. At equilibrium, the flow of electrons from the semi-conductor to metal is balanced by the contact potential so that there is no net current. While biasing a junction, the system remains no longer in equilibrium as electrons or holes have been injected in to the system by applying external potential. So, in bias the Fermi levels will no longer line up but, will be shifted depending upon the applied potential. By applying an external potential, the Schottky junction can be biased in two ways

1. Forward bias - metal is connected to positive terminal and n-type semi-conductor connected to negative terminal
2. Reverse bias - metal is connected to negative terminal and n-type semi-conductor connected to positive terminal

The current flow depends on the type of bias and the amount of applied external potential.

### 2.11.1.1 Forward bias

In a forward biased Schottky junction, the metal is connected to the positive side and the semiconductor is connected to the negative side of an external potential. We find that external potential is applied in such a way that it is opposite to the contact potential that is setup within the junction. Since the region with the highest resistivity is the depletion region near the junction, the voltage drop takes place across the depletion region. As already mentioned

above, under external bias, the Fermi levels no longer line up, but are shifted with respect to one another and the magnitude of the shift depends on the applied voltage. Energy band diagram of the Schottky junction under forward bias is shown in figure 9. Thus, electrons injected from the external circuit into the n-type semiconductor have a lower barrier to overcome before reaching the metal. This leads to a net current in the circuit which further increases with the increase in external potential. The current in a Schottky diode under forward bias is given by

$$J = J_0 \left[ \exp \left( \frac{eV}{K_B T} \right) \right] \quad (6)$$

where  $J$  is the current density for an applied potential of  $V$ .  $J_0$  is a constant and depends on the Schottky barrier ( $\phi_B$ ) for the system and the expression is

$$J_0 = \left[ \exp \left( \frac{-\phi_B}{K_B T} \right) \right] \quad (7)$$

where  $A$  is the Richardson constant for thermionic emission and is a material property. Equation 6 shows that in the forward bias the current depends exponentially on the applied potential and the higher the potential, higher will be the current

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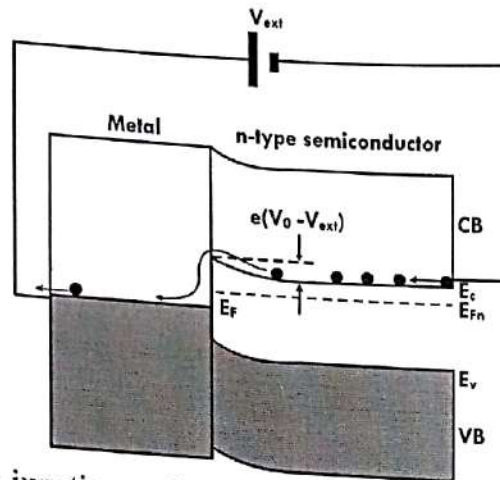


Figure 9: Schottky junction under forward bias. Adapted from Principles of Electronic Materials - S.O. Kasap.

### 2.11.1.2 Reverse bias

In the case of a reverse bias, the metal side is connected to a negative and the semiconductor is connected to a positive. In this particular case, the external potential  $V$  is in the same direction as the contact potential, which means the overall barrier goes up for electron motion from the n-type semiconductor to metal. There is a small current that goes from the metal to the semiconductor and this reverse current depends upon the Schottky barrier ( $\phi_B$ ). So there is a constant current in reverse bias, whose magnitude is equal to  $J_0$  (as given in equation 7). From calculations it can be shown that the current in the forward bias is orders of magnitude higher than the current in reverse bias (this arises due to the exponential dependence on potential). So a Schottky junction acts as a rectifier i.e. it conducts in forward bias but not in reverse bias. The I-V characteristics of the junction are shown in figure 11. There is an exponential increase in current in the forward bias (I quadrant) while there is a small current in reverse bias (IV quadrant).

*Schottky Rectifier in FB*



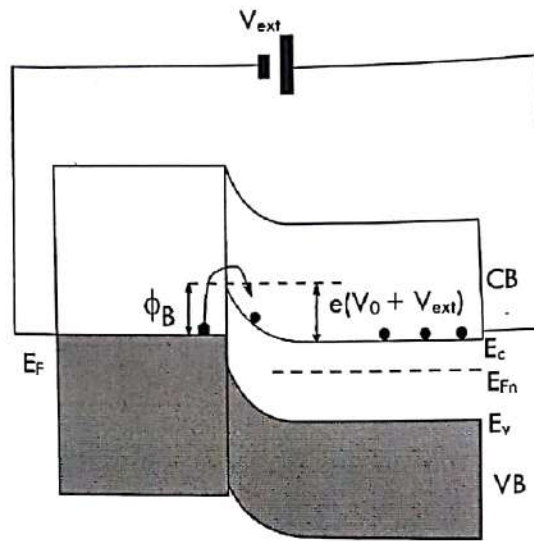


Figure 10: Schottky junction under reverse bias. Adapted from Principles of Electronic Materials - S.O. Kasap.

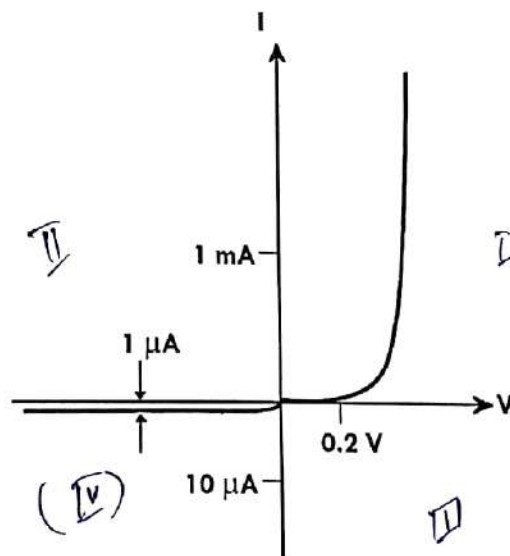


Figure 11: I V characteristics of a Schottky junction showing rectifying properties. Adapted from Principles of Electronic Materials - S.O. Kasap.

### 2.11.2 Ohmic junction

$\phi_m < \phi_{semi}$   
Fermi of metal  $>$  Fermi of semi

A Schottky junction is formed when the semiconductor has a lower work function than the metal. When the work function of the metal is smaller than the work function of the semiconductor, the junction formed is called the Ohmic junction. The energy band diagram of the junction in equilibrium (Fermi levels line up) is shown in Figure 13. There is large number of electrons in the metal side and these electrons can move to the semiconductor side. So, an electric field will again be setup and the field will be in the direction opposite to that of a Schottky Junction. As electrons are going from the metal to the empty states in the conduction band, so, there is an accumulation region near the interface (on the semiconductor side). The accumulation region has a higher conductivity than the bulk of the semiconductor due to this higher concentration of electrons. Thus, a Ohmic junction behaves as a resistor conducting in both forward and reverse bias. The resistivity is determined by the bulk resistivity of the semiconductor.

One of the interesting applications of Ohmic junctions is in thermoelectric devices, where a small volume can be cooled by application of direct currents. For Ohmic junctions, depending on the direction of current flow (forward or reverse bias), heat can be generated or absorbed. This is shown in figure 14. This is called Peltier effect and is applicable to junctions between any two dissimilar materials. So, when current flows through a metal semiconductor Ohmic junction, heat will always be released or absorbed. This can be used as a practical cooling device, as shown in the schematic in figure 15. Both p and n type semiconductors are used and the current flow is such that one end is always cold (heat absorbed) while the other end is hot (heat released).

Ohmic as Resistor in FB & RB

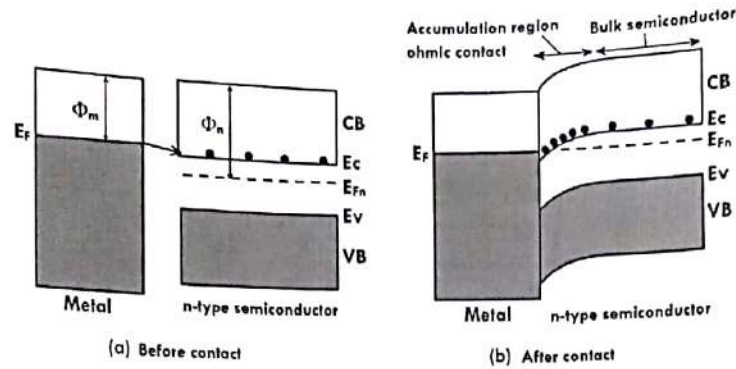


Figure 13: Ohmic junction (a) before and (b) after contact. Before contacts the Fermi levels are at different positions and they line up on contact to give an accumulation region in the semiconductor. Adapted from Principles of Electronic Materials - S.O. Kasap.

There are different metals that form Ohmic contacts with semiconductors. A partial list of such combinations is shown in table 1. This list is important because it gives the acceptable metals for each semiconductor that can be used in fabrication of devices. For Si, earlier Al was used for making electrical contacts, which was then replaced by Al-Cu alloys. Both form Ohmic contacts with Si. These have now been replaced by Cu (higher conductivity). But, Cu is poisonous to Si and hence cannot directly come into contact with the semiconductor. Usually, Ti or TiN is used to make electrical contact with Si, and then Cu is deposited. Ti or TiN is chosen because it also forms Ohmic contacts with p and n type Si, see table 2, and can be easily deposited by chemical vapor deposition.



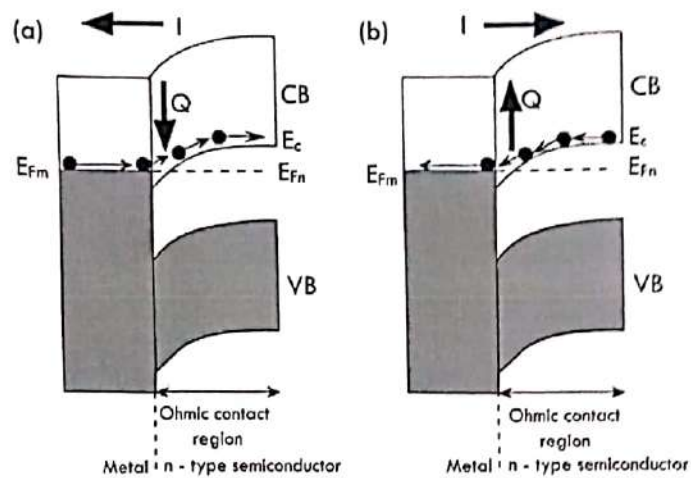


Figure 14: Current flow through an Ohmic junction can lead to heat (a) absorption or (b) release. This depends on the external bias, that determines the direction of heat ow. When electrons move from metal to higher energy levels in the semiconductor heat is absorbed and the reverse happens when electrons ow from semiconductor to metal. Adapted from Principles of Electronic Materials - S.O. Kasap.

compul

Unit - 2

Topic - 7

① Gigan

## Semiconducting Materials for optoelectronic

Study of semiconductor materials began in the early 19th century

### ① Elemental Semiconductors :-

The elemental semiconductors are those composed of single species of atoms such as Si, Ge, Sn in column IV and Selenium (Se) & Tellurium (Te) in column VI of periodic table.

### ② Compound Semiconductors :-

There are numerous compound semiconductors, which are composed of two or more elements

\* GaAs (Gallium Arsenide) is a binary III-V compound which is a combination of gallium (Ga) from column III and Arsenic (As) from column V.

\* Ternary compounds can be formed by elements from three different columns e.g. Mercury Indium Telluride ( $\text{HgIn}_2\text{Te}_4$ ) a II-III-VI compound.

They can also be formed by elements from two columns such as aluminum gallium Arsenide ( $\text{Al}_x\text{Ga}_{1-x}\text{As}$ ) which is a ternary III-V compound where Al and Ga are from III and subscript  $x$  is related to composition of two elements from 100 percent Al ( $x=1$ ) to 100 percent Ga ( $x=0$ ).

Pure Si is most imp. material for integrated circuit applications, and III-V binary & ternary compounds are most significant for light emission.



## for Diode Lasers? -

### (a) III-V Semiconductors! →

As discussed earlier, ~~the~~ III-V columns of periodic tables in the form of  $A_x B_{1-x} C$  where  $x$  - mole fraction in the range from 0 to 1, <sup>have</sup> emerged as the materials of choice for lasers that emit in the  $(\lambda = 0.7 - 1.6 \mu\text{m})$  range of wavelength.

This wavelength range includes

- (i) fibre optic communication bands ( $0.85, 1.31 \text{ \& } 1.55 \mu\text{m}$ )
- (ii) pumping bands for fibre Amplifiers ( $1.48 \text{ \& } 0.98 \mu\text{m}$ )
- (iii) window for pumping Nd-YAG laser ( $0.81 \mu\text{m}$ )
- (iv) wavelength currently used for optical disk players ( $0.78 \mu\text{m}$ )

there are three Group three elements (Al, Ga, In)

d. Three group five atoms (P, As, Sb) to

there are 18 possible ternary III-V solutions. Some of III-V solid solutions have direct band gaps & others have not. So, not all ternary semiconductors are good optical materials.

(b) In addition to usual III-V compounds, some nitride compounds for e.g. AlN (Hexagonal) have gained attention because of successful demonstration of LEDs emitting at high energies in the visible spectrum. + (d) part combine + (h) part combine

(c) IV-VI semiconductors? → \* Tunable semiconductor light sources based on these materials are useful for high resolution spectroscopy applications.

\* Most light emitting IV-VI semiconductors devices operate at cryogenic temp. typically at 80K. For e.g.  $\text{Pb}_{1-x}\text{Sn}_x\text{Te}$ ,  $\text{Pb}_{1-x}\text{Sn}_x\text{S}$ ,  $\text{Pb}_{1-x}\text{Sn}_{1-x}\text{Se}$ ,  $\text{Pb}_{1-x}\text{Sn}_{1-x}\text{Te}$



Direct

(III-V)

(d) Gallium Nitride (GaN) & Indium Gallium Nitride are used in semiconductor laser with blue beams ( $\lambda \sim 445 \text{ nm}$ ). GaN transistors can operate at high voltages and high temp than GaAs.

Indirect

(e) Gallium phosphide (III-V) is used in early low to medium brightness cheap red/orange/green LEDs. Pure GaP emits green, nitrogen doped GaP emits yellow-green. ZnO doped GaP emits red.

Direct

(f) GaAs (III-V)  $\rightarrow$  It is second most common in use after Si and is commonly used as substrate for other III-V semiconductors e.g. InGaAs & GaInNAs. It is used for Near-IR LEDs, fast electronics & high efficiency solar cells.

Direct

(g) GaSb (III-V)  $\rightarrow$  It is used for IR detectors & LEDs with thermophotovoltaics. It is commonly doped n with Te, & p with Zinc.

Indirect

(h) Indium Nitride (InN) is commonly used in solar cells. It is used frequently as alloys as p-type doping is difficult in this.

Direct

(i) III-V (Indium Arsenide) (InSb) is used in IR detectors and thermal imaging sensors. AlInSb-InAlInSb structures are used as quantum well.

(j)

II-VI

(CdSe) - Cadmium selenide are tested for high efficiency solar cells. Similarly CdS is used in photoresistors and solar cells (also with CdTe). CdTe (cadmium telluride) is also used in photovoltaics & electro-optic modulators.





(K) II-VI (ZnSe) Zinc selenide is used for blue lasers and LEDs. It is the most common material used in infrared optics. ZnTe (Zinc Telluride) can be grown on AlSb, GaSb, InAs, PbSe and is used in solar cells, blue LEDs and lasers. Together with Lithium Niobate, it is used to generate terahertz radiation.

(L) (IV - VI)  $\rightarrow$  \* (Tin & Sulphide)  $\rightarrow$  (SnS<sub>2</sub>) is used in gas sensing applications.

\* Lead Tin telluride (PbSnTe)  $\rightarrow$  used in IR detectors & for thermal imaging.

(M) ~~Copper~~ Copper tin sulphide (Cu<sub>2</sub>SnS<sub>3</sub>) is p-type semiconductor & is used in thin-film solar application.

(N) Typical laser semiconducting materials.

\* GaAs based lasers are used in near IR region.

\* PbX (X is chalcogenide like S, Se, Te) lasers are used for mid IR region (larger wavelengths).

\* CdX (X is chalcogenide) lasers are used in visible region.

Numerical :-

\* Electron conc. in s-c varies linearly from a value

$1 \times 10^{24} \text{ per m}^3$  to  $6 \times 10^{23} \text{ per m}^3$  over a distance of 0.1 cm.

If diffusion constant is  $0.025 \text{ m}^2/\text{s}$ , find diffusion current density in sample.

Soln:  $\frac{dn}{dx} = \frac{10^{24} - 6 \times 10^{23}}{0.001} = 4 \times 10^{26} \text{ m}^{-4}$

Diffusion current density :-  $j = e D_n \frac{dn}{dx}$   
 $= 1.6 \times 10^{-19} \times 0.025 \times 4 \times 10^{26}$   
 $= 1.6 \times 10^6 \text{ A/m}^2$